

NOKOLSKI, AK, USA

WMO: 998429

Lat: 52.941N

Lon: 168.871W

Elev: 16

StdP: 14.69

Time Zone: -9.00 (NAK)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.9	24.7	N/A	N/A	N/A	N/A	N/A	N/A	56.2	32.4	52.1	35.6	16.0	310	0.947

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.6	57.1	N/A	55.3	N/A	54.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	16.2	140

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Extreme Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
47.5	41.9	36.7	15.8	60.2	5.0	2.8	12.2	62.3	9.3	63.9	6.5	65.5	2.8	67.6		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.2	33.9	34.5	35.0	37.0	41.1	44.9	48.5	50.5	48.8	44.0	39.2	36.6
	DBStd	6.69	4.74	4.34	3.93	2.92	2.47	2.04	2.20	2.52	3.25	3.09	3.56	4.44
	HDD50	3289	498	433	466	390	277	154	56	25	59	190	324	417
	HDD65	8688	963	853	931	840	742	604	511	450	486	652	774	882
	CDD50	74	0	0	0	0	0	0	10	39	23	2	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	16.6	18.5	18.8	15.8	17.0	15.4	14.5	13.9	13.4	16.0	19.0	19.1	18.6
Precipitation	PrecAvg	53.5	6.0	5.5	4.7	2.8	3.1	2.6	3.5	4.0	4.5	4.9	5.6	6.4
	PrecMax	67.2	10.5	7.9	9.9	4.4	5.2	4.0	5.4	5.9	6.7	8.4	10.9	10.9
	PrecMin	39.8	2.7	1.9	1.6	1.5	1.5	1.1	0.9	1.9	1.7	3.4	3.0	2.7
	PrecStd	6.6	1.7	1.4	1.9	0.7	0.8	0.7	1.0	1.0	1.3	1.0	1.8	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	42.9	43.4	45.9	49.1	52.3	56.2	59.3	61.4	54.2	49.5	47.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	42.3	41.6	42.2	43.3	46.7	50.3	54.5	56.9	58.4	51.6	47.3	45.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	41.2	40.8	41.1	42.1	45.4	49.2	53.2	55.5	55.1	49.9	45.7	43.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	40.0	39.9	40.0	41.2	44.3	48.1	52.0	54.6	53.2	48.4	44.3	42.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	5.9	5.8	6.3	5.8	5.5	5.1	4.8	5.6	5.9	5.6	6.4	6.0
		MCDBR	5.4	5.8	6.9	7.0	6.9	7.0	6.5	6.9	7.6	6.7	7.3	6.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.321	0.348	0.361	0.417	0.407	0.402	0.404	0.390	0.382	0.355	0.338	0.309	
	taud	2.257	2.255	2.274	2.115	2.223	2.323	2.383	2.425	2.337	2.302	2.205	2.245	
	Ebn at Noon	203	235	259	257	267	270	267	265	250	229	190	183	
	Edh at Noon	21	28	33	44	42	38	36	32	31	26	21	17	
All-Sky Solar Radiation	RadAvg	199	398	757	1117	1269	1282	1129	1015	849	542	258	151	
	RadStd	17	24	72	108	119	111	97	74	60	36	21	15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	+2.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page